

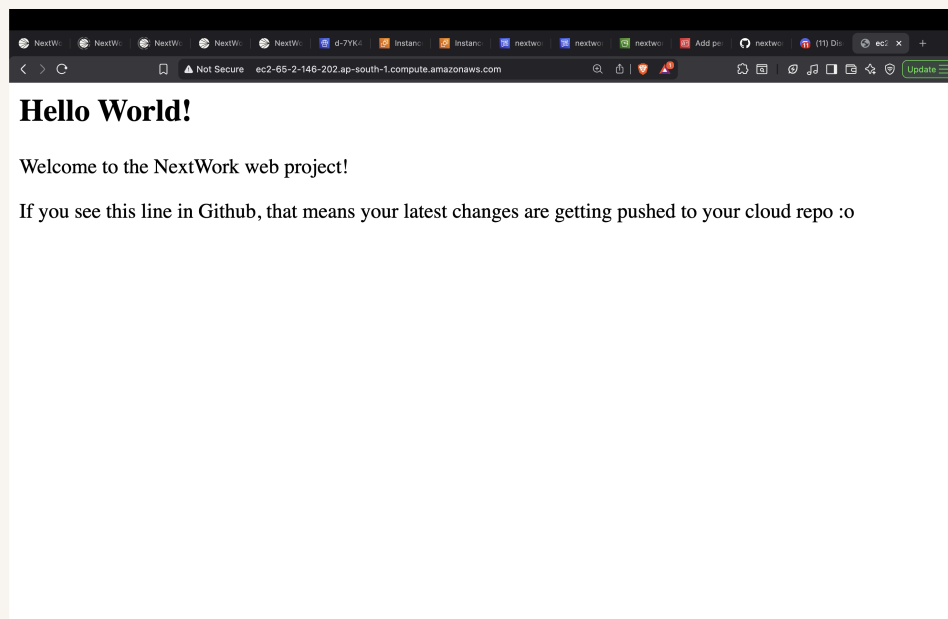


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Deploy a Web App with CodeDeploy



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Introducing Today's Project!

In this project, I will demonstrate how to use CodeDeploy to deploy a web application. I'm doing this project to learn about how deployment works, and how it can be automated using a combination of CodeDeploy and deployment. By the end of this project, you can get see a live website.

Key tools and concepts

Services I used were EC2, IAM, CodeArtifact, CodeBuild, CloudFormation, S3, CodeConnection and CodeDeploy Key concepts I learnt include deployment, deployment groups, deployment scripts, appspec.yml and importance of rebuilding my project before i deploy.

Project reflection

This project took me approximately 4 hours including building and troubleshooting. The most challenging part was CodeDeploy where i had mistaken in configuring things in CodeBuild creation in which i had to name the build artifact to the created bundle for deployment but i leave the name by enabling semantic version(it does not need name) inside the S3 bucket directory, as CodeDeploy will search for zip file bundle in S3 bucket but it doesn't find any with name zip file, so i have to edit CodeBuild and rebuild. It was most rewarding to see the deployed web application.



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This project is part five of a series of DevOps projects where I'm building a CI/CD pipeline!



Deployment Environment

To set up for CodeDeploy, I launched an EC2 instance and VPC because EC2 instance will serve as the virtual server where my web application will be hosted and run. In other terms one EC2 is for development and other for deployment. The VPC provides a secure and isolated network environment in AWS where your resources, like EC2 instance, can run without being exposed to the public internet unless I specifically configure them to be.

Instead of launching these resources manually, I used CloudFormation. When I need to delete these resources created by CloudFormation Template, I can simply delete an entire stack.

Other resources created in this template include networking resources like VPC's, Internet Gateways, Security Group's, EC2 instance, Subnet and Route Tables. They're also in the template because production environment has specific requirements around traffic that it should enable and block out. Setting up all of the resources that will help deployment work as common practice.



CloudFormation > Stacks > NextWorkCodeDeployEC2Stack

Stacks (1)  

Filter status: Active 

View nested 

Stacks

- NextWorkCodeDeployEC2Stack
2026-02-25 23:32:10 UTC+0530
 CREATE_COMPLETE

NextWorkCodeDeployEC2Stack

Delete stack Update stack Stack actions Create stack

Stack info Events **Resources** Outputs Parameters Template Changesets Git sync

Resources (11) 

Search resources

Logical ID	Physical ID	Type	Status	Module
DeployRoleProfile	NextWorkCodeDeployEC2Stack-DeployRoleProfile-rj6mffWVYlBku	AWS::IAM::InstanceProfile	 CREATE_COMPLETE	-
InternetGateway	igw-01e7cd3326bc71803 	AWS::EC2::InternetGateway	 CREATE_COMPLETE	-
PublicInternetRoute	rtb-0265dd916c7adb686j0.0.0.0/0	AWS::EC2::Route	 CREATE_COMPLETE	-
PublicRouteTable	rtb-0265dd916c7adb686	AWS::EC2::RouteTable	 CREATE_COMPLETE	-
PublicSecurityGroup	sg-002938f94cb672ff8 	AWS::EC2::SecurityGroup	 CREATE_COMPLETE	-
PublicSubnetA	subnet-073e1bc29310dd0e4 	AWS::EC2::Subnet	 CREATE_COMPLETE	-



Deployment Scripts

Scripts are many programs that helps us to automate the running of commands. To set up CodeDeploy, I also wrote scripts to automate deployment commands i.e., these are commands that my deployment EC2 instance needs to run inorder to post to my web application.

The 'install_dependencies will help my EC2 instance install all dependencies it needs to host my web app like Tomcat and Apache.

The start_server.sh will starts up two servers Apache and Tomcat needed in EC2 instace if in case it is rebooted. The scripts written in this file is like start and enable the Tomcat, that responsible for running the java app and start and enable apache the web server that is responsible handling web traffic and passing requests to Tomcat.

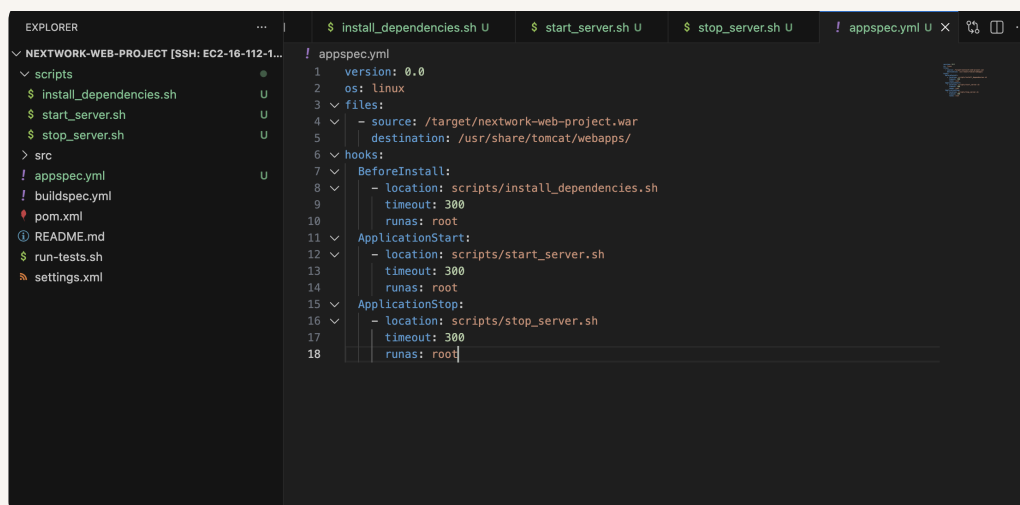
The stop_server.sh will scripts that stops my Apache and Tomcat servers where they are no longer needed to serve the web app to a user.



appspec.yml

Then, I wrote an appspec.yml file to give CodeDeploy the instructions for deploying my web app. The key sections in appspec.yml are BeforeInstall (i.e., use the install_dependencies.sh scripts), ApplicationStart (i.e., use the start_server.sh script), and ApplicationStop (i.e., use the stop-server.sh script).

I also updated buildspec.yml because to tell CodeBuild it should also to package up the new appspec.yml and setup scripts that i created inside the build artifact(i.e., the compressed war file).



```
! appspec.yml
1  version: 0.0
2  os: linux
3  files:
4  - source: /target/nextwork-web-project.war
5    destination: /usr/share/tomcat/webapps/
6  hooks:
7  - BeforeInstall:
8    - location: scripts/install_dependencies.sh
9      timeout: 300
10     runas: root
11  - ApplicationStart:
12    - location: scripts/start_server.sh
13      timeout: 300
14     runas: root
15  - ApplicationStop:
16    - location: scripts/stop_server.sh
17      timeout: 300
18     runas: root
```



Setting Up CodeDeploy

A deployment group is group of EC2 instances that i deploy to, and also the collections of settings that determine how i want to deploy my web application. A CodeDeploy application is simply a folder that holds them together all the deployment groups for the same application.

To set up a deployment group, you also need to create an IAM role to give CodeDeploy the permission to access the EC2 instances that it needs to coordinate. Otherwise CodeDeploy doesn't have access to EC2 that equals to not being able to give my deployment instances the instructions to deploy the web application.

Tags are helpful for identifying the EC2 instances that will be deploying the web application. I used the tag 'role:webserver' to automatically match EC2 instance i deployed with my CloudFormation template. Now at the future, when i want to have other instances also wanted to be in this deployment group, i can simply tag 'role:webserver' tag. They'll automatically get added upto the deployment group.



Tag group 1

Key	Value - optional	
role	webserver	<button>Remove tag</button>
<button>Add tag</button>		
<button>+ Add tag group</button>		

☐ On-premises instances

Matching instances

1 unique matched instance. [Click here for details](#)



Deployment configurations

Another key settings is the deployment configuration, which affects how quickly and risky we are deploying the web application. I used `CodeDeployDefault.AllAtOnce`, so any changes that CodeDeploy deploys the affects the all the instances in the deployment group at once. This is fastest wat to deploy a change/code update.

In order to connect the deployment instance with CodeDeploy. A CodeDeploy Agent is also set up to recieve instructions from CodeDeploy, and make sure that the commands in `appspec.yml` are run.

Agent configuration with AWS Systems Manager info

❗

Complete the required prerequisites before AWS Systems Manager can install the CodeDeploy Agent.
Make sure that the AWS Systems Manager Agent is installed on all instances and attach the required IAM policies to them. [Learn more](#)

Install AWS CodeDeploy Agent

☐ Never

☐ Only once

☒ Now and schedule updates

Basic scheduler

Cron expression

14

Days

▼



Success!

A CodeDeploy deployment is a specific update to my application that i am deploying to users. The difference to a deployment group is that group is like the settings file and the deployment itself is a specific update we roll out using that settings file.

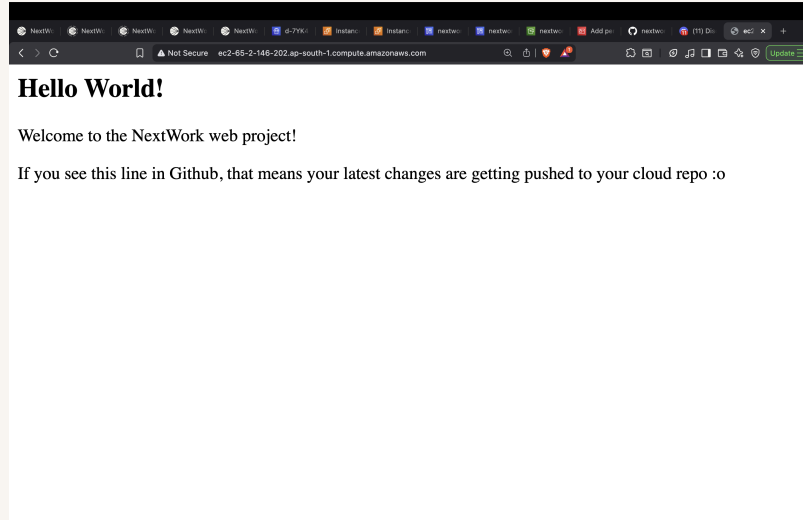
I had to configure a revision location, which means where my web application's WAR file(i.e., the compressed file that is ready to be deployed) lives. My revision location was the S3 bucket that i've created and linked with CodeBuild (i.e., build artifacts that will automatically goes into that bucket).

To check that the deployment was a success, I visited the IPV4 DNS address of the deployment EC2 instance. I saw a live web app that's working and serving my web app code to end users (which is my devops project objective).



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Disaster Recovery

In a project extension, I decided to create a broken deployment by means of leaving an intentional error in the code to demonstrate rollback for the disaster recovery strategy. The intentional error I created was updating `stop_server.sh` file as `systemctl` (i.e., adding an extra 'l' in the command making it to unrecognizable command that will error). This will cause the deployment to fail because the command `systemctl` is not a command thus leading it to a "command not found" error.

I also enabled rollbacks with this deployment, which means it lets me to go to previous version which was working fine, if something in the current version goes wrong. It is enabled in most of the current projects because if running applications breaks during deployment, it rolls back to previous working condition in minutes.

'When my deployment failed, the automatic rollback is also failed because Rollback in deployment actually rolls back the previous configuration that worked fine. Now with previous configuration point out the same build Artifact that was built with error file of CodeBuild, this deployment also fails. To actually recover from this failed deployment, I'd have to manually change code to correct code and push the changes, Rebuild the CodeBuild and then Start new deployment. In production environments, automated Rollbacks are often implemented with more advanced tools. AWS CodePipeline can be configured to automatically rollback to the last successful deployment when a failure is detected, which eliminates the need of manual intervention.



Developer Tools > CodeDeploy > Deployments ⓘ

Deployment history

[View details](#) [Actions](#) [Copy deployment](#) [Retry deployment](#)

< 1 > ⚙

	Deployment...	Status	Deploym...	Compute ...	Application	Deploym...	Revision L...	Initiating event	Start time	End time
☐	d-YA4616OZG	Failed	In place	EC2/on-pr...	nextwork-...	nextwork-...	s3://next...	CodeDeploy rollback	Mar 3, 2026 ...	Mar 3, 2026 10:01 P...
☐	d-I7CJRUNZG	Failed	In place	EC2/on-pr...	nextwork-...	nextwork-...	s3://next...	User action	Mar 3, 2026 ...	Mar 3, 2026 10:01 P...



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